Week 9 – Workshop
Geothermics: Transient Heat Flow

Geophysics 3A: Potential Fields & Geothermics

July 31, 2026

Preparation

Pre-reading: Course Notes, Ch. 11
Lecture videos:
Notes:

Purpose

This workshop develops intuition for transient heat flow: how thermal diffusivity sets the timescales and length scales over which thermal signals propagate, how half-space and plate cooling models describe the thermal evolution of oceanic lithosphere, and how latent heat modifies cooling in Stefan problems.

Learning Outcomes

Students will:

- Understand how thermal diffusivity controls timescales.
- Compare thermal and chemical diffusion processes.
- Interpret half-space and plate cooling models.
- Use Fourier series to conceptualize thermal evolution.

Session Flow (⁇ min)

0–10 min: Warm-up 9.A: Wedge vs. Orogen

Both accretionary wedges and orogens form at convergent margins, but they experience very different thermal histories. Before we build a detailed model for one of them later today, predict how surface heat flow evolves over the life of each.

10–25 min: Mini-lecture — The diffusion equation and thermal diffusivity

- The heat (diffusion) equation: $\partial T/\partial t = \kappa \nabla^2 T$.
- Thermal diffusivity $\kappa = k/(\rho c_p)$: what controls it, typical values for crustal rocks.
- Scaling argument: thermal length $z \sim \sqrt{\kappa t}$; thermal timescale $t \sim z^2/\kappa$.
- Key difference from steady state: temperature *evolves* in time.

25–65 min: Activity 9.B: Transient Geotherms

Tectonic and other geologic events often have a thermal effect on the lithosphere, whether top, bottom, or internally. In this activity, you'll explore several scenarios related and learn how to approximate their temporal evolution intuitively.

65–110 min: Activity 9.C: Thermal Diffusivity, Thermal Length, and Timescales

- Review $\kappa = k/(\rho c_p)$.
- Emphasize modest variation in crustal rocks.
- Contrast sediments, rock, and ice.
- Penetration depth scaling: $z \sim \sqrt{\kappa t}$.
- Borehole paleothermometry vs ice-core records.
- Depth–resolution tradeoffs.

110–120 min: Break

120–155 min: Activity 9.D: Oceanic Lithosphere: Half-Space and Plate Cooling

- Derive surface heat flow and seafloor subsidence from the half-space cooling solution.
- Test both predictions against real global heat flow and bathymetry data.
- Compare to the plate cooling model to see where and why the half-space prediction breaks down, at $t \sim \tau = L^2/\kappa$.

155–165 min: Wrap-up

What single quantity controls how fast heat moves through rock, and what are its typical values? Nearly all thermal problems result in a time or distance scaling that comes from the argument of an error function or exponential similar to $t \sim z^2/\kappa$. Can you give one geological example from each activity where this scaling is evident?

165–175 min: Preview: Finite Differences

Analytic solutions to the diffusion equation require simplifying assumptions: constant properties, simple geometries, and idealized boundary conditions. Real geological systems violate all of these Next week we develop finite difference methods, which replace the differential equation with a local energy balance that can be applied to arbitrarily complex geometries and boundary conditions. You will first derive the finite difference equations from first principles and then solve the heat equation yourself by hand—becoming the computer.

Workshop Notes

Workshop Notes

Warm-up 9.A
Wedge vs. Orogen

Purpose

Both accretionary wedges and orogens form at convergent margins, but they experience very different thermal histories. Before we build a detailed model for one of them later today, predict how surface heat flow evolves over the life of each.

Accretionary wedge

Sediment and crust are continuously scraped off a subducting oceanic plate and stacked at the trench. The wedge is built from material that has spent little time near the surface; it arrives cold, wet, and is rapidly buried by whatever is scraped off next.

- While subduction is active, is heat flow through the wedge likely to be higher or lower than a stable craton? What is actually supplying heat to this material, and what is working against it staying warm?

- Once subduction stops—no more cold material arriving, no more rapid burial—what happens to heat flow? Does it recover to the same level as before subduction started, or could it overshoot?

Orogen

Continental collision thickens the crust; doubling it is not unusual. The thickened crust contains more total radiogenic material than before, buried deeper than it used to be.

- Right after thickening, is the crust hotter or colder than it will eventually become? *Hint:* think about how long it takes buried heat production to actually raise temperatures at depth.

- Once the orogen has had a long time to adjust, how does its eventual heat flow compare to a stable craton: higher, lower, or about the same?

Sketch both

On the axes below, sketch your predicted heat flow through time for *both* settings, on the same plot. The dashed line marks typical cratonic heat flow, for reference.

We will not resolve the wedge case in detail today – keep your prediction in mind as a point of contrast. The orogen case is looked at in greater detail in scenarios examined later.

Activity 9.B
Transient Geotherms

Purpose

A steady-state geotherm is a snapshot at the end of a very long evolution. This activity asks what happens in between: given a geological event that perturbs a crust already at steady state, sketch the geotherm immediately after the event, reason about where it is heading, and sketch how it gets there.

Learning Outcomes

By the end of this activity, you should be able to:

- Sketch the geotherm immediately after a perturbing event, given the pre-event geotherm and a description of what the event does to the crust.
- Use the curvature of a geotherm to predict where and how fast it will change, and recognize when heat production, latent heat release, or a changing boundary condition breaks that simple rule.
- Reason from a scenario's boundary conditions alone whether its new equilibrium surface heat flow should be higher, lower, or unchanged.
- Connect a scenario's evolving geotherm to its surface heat flow (from the gradient at $z = 0$) and subsidence (from the integrated temperature difference), and sketch both through time.

Box .1: Theory

Reading Curvature

The heat equation is

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial z^2} + \frac{A}{k} \right).$$

With no heat production ($A = 0$), this reduces to

$$\frac{\partial T}{\partial t} \propto \frac{\partial^2 T}{\partial z^2};$$

temperature changes fastest exactly where the geotherm is most sharply curved, and does not change at all where it is a straight line. A kink, a sudden change in slope, is a point of enormous curvature, so it smooths out very quickly relative to everything else on the curve. With heat production ($A \neq 0$), there is a persistent curvature, exactly $-A/k$, even once equilibrium has been reached.

Derived quantities

Once a geotherm is known, the surface heat flow can be determined by Fourier's Law, i.e., differentiation of temperature evaluated at the surface,

$$q_s(t) = k \left. \frac{\partial T(t)}{\partial z} \right|_{z=0}.$$

Temperature changes have the potential to affect the lithospheric buoyancy. Isostatic change in elevation is an integrated difference between the geotherm and a reference:

$$\Delta\varepsilon(t) = \alpha \int_0^{z_{min}} [T(t) - T_{\text{ref}}]\, \mathrm{d}z,$$

Box .1: continued.

where z_{min} is the depth at which the geotherms converge. We often refer to the subsidence as most thermal processes result in lower elevation as the system cools. Subsidence has the opposite sign as elevation.

The Scenarios

In each of the scenarios below, there are two associated subplots. The bottom subplot includes the geotherm prior to the perturbing event. This geotherm is in steady-state and includes a modest amount of heat production. On this subplot you will draw the perturbed geotherm, the equilibrium geotherm and several steps detailing the temporal evolution of the geotherm.

The top axes are for plotting the change in heat flow q and subsidence, s, throughout the evolution of the event. The horizontal, t-axis represents the heat flow and subsidence prior to the initiating event.

In the description of each scenario below includes:

- the effect of the event on the crust;
- the change in boundary conditions resulting from the event; and
- the equilibrium condition if it cannot be easily reasoned.

To make things easier, we assume the initiating event is instantaneous. This assumption is common when developing analytic transient solutions. Numerical methods (Week 10) let us relax this assumption and incorporate more complex, time-dependent evolution.

A. Sedimentation

A package of sediments has been deposited on the crust with thickness, h. The surface boundary temperature remains unchanged. Deep in the crust, the heat flow entering from the mantle is unaffected by what happens above it; take this as fixed at its pre-event value.

B. Erosion

The upper part of the crust has been eroded by h kilometers. The temperature at the surface has remained constant at T_s throughout the process. Like sedimentation, the deep crust is unaffected by the heat flow entering from the mantle.

C. Thrust

During an orogenic event, the upper h kilometers of crust has been duplicated. The mantle heat flow entering the base of the crust is unchanged from its pre-event value. *Note:* neglecting any other processes that occur during the orogeny, consider what affect this has on the equilibrium surface heat flow.

D. Magmatic Intrusion

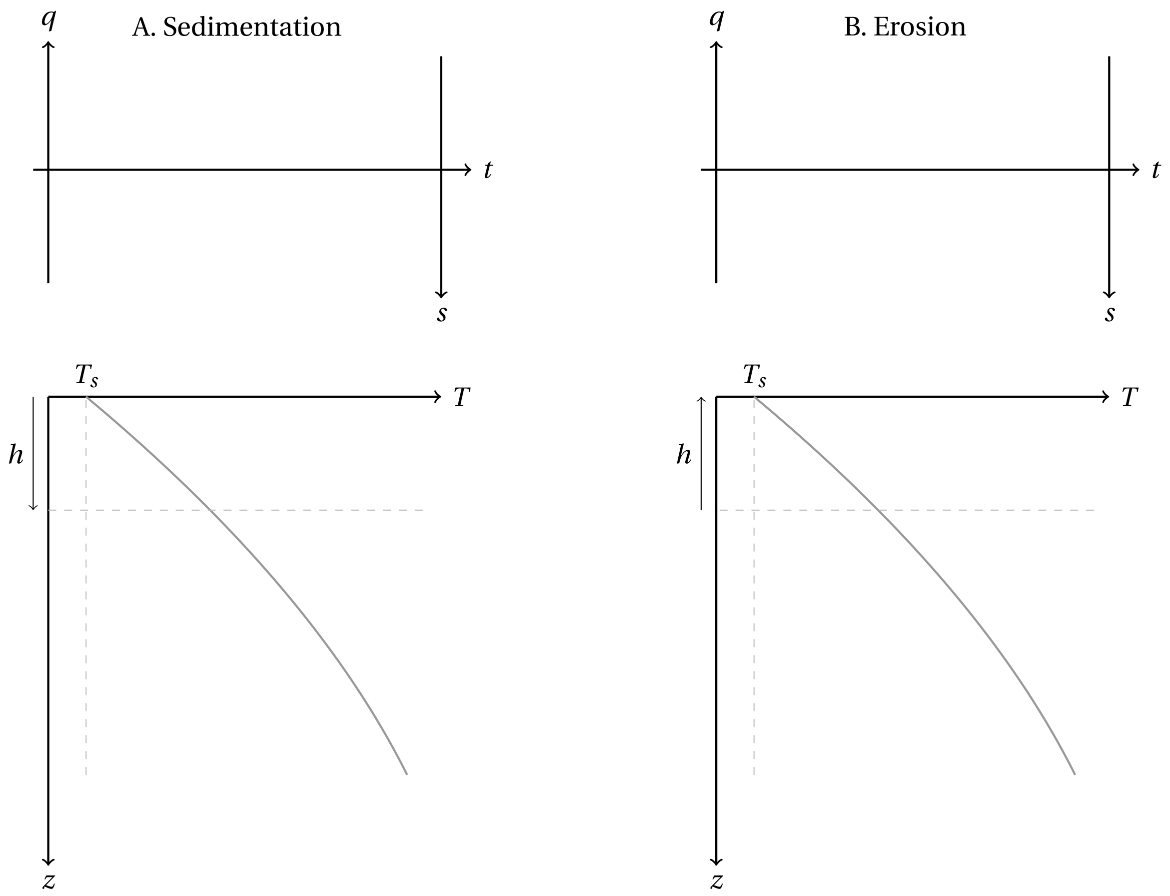
A magma body of thickness h and temperature T_i has been intruded into the upper crust. The surface and mantle boundary conditions remain unchanged. *Note:* latent heat releases progressively as the body crystallizes, not all at once.

E. Lower Crustal Delamination

A lower crust delamination occurs in response to a large density increase related to melt extraction and subsequent metamorphism. The lower h kilometers of crust has foundered, bringing the mantle temperature T_m into contact with the new base of the crust. As a result, the mantle heat flow has increased as convecting mantle has moved to fill the gap.

F. Deglaciation

Following a deglaciation, the temperature at the top of the solid Earth increases from T_g to T_s. No ice is depicted in the pre-event geotherm.



Question 1. Sedimentation vs. Erosion

(a) On the Sedimentation panel, sketch the geotherm immediately after deposition.

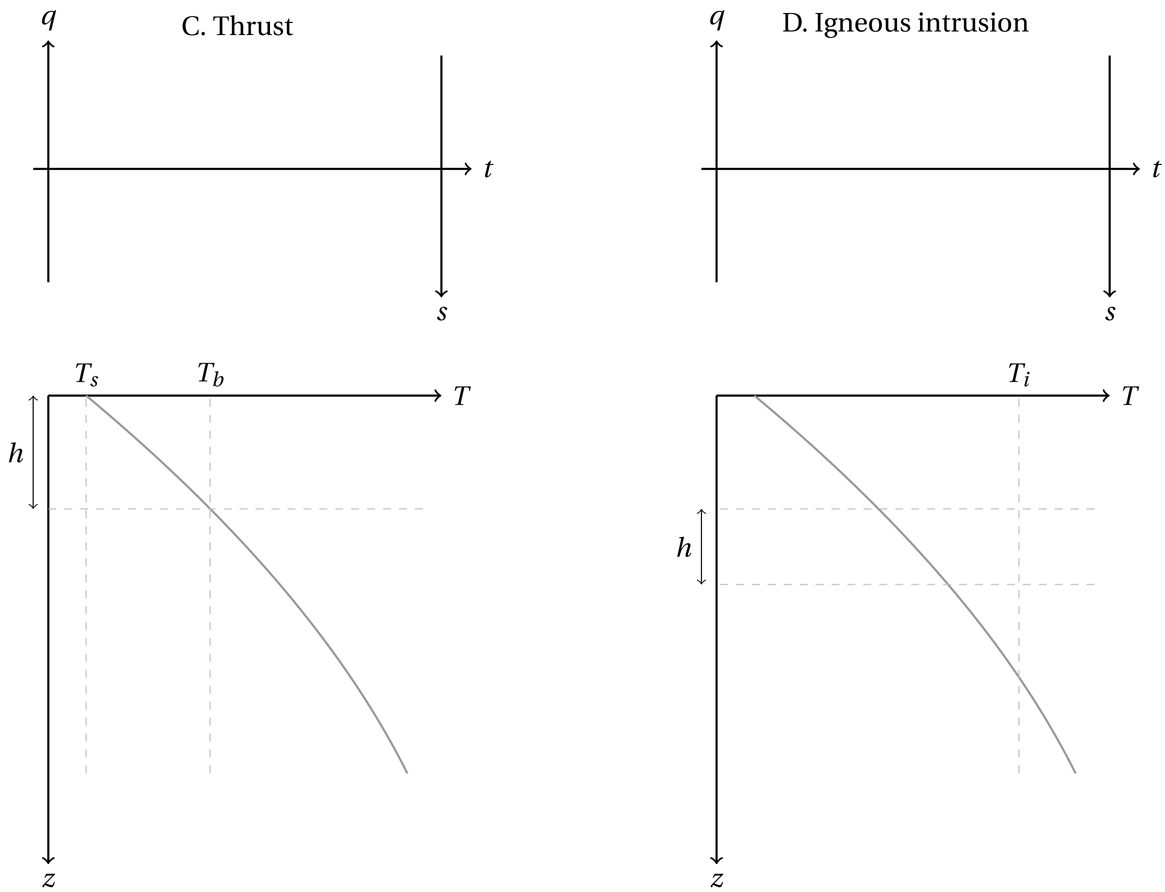
(b) On the Erosion panel, sketch the geotherm immediately after erosion.

(c) In Sedimentation, is the resulting discontinuity a kink *within* the profile, or a jump *at* the surface? What about Erosion? Mark the point of highest curvature on each sketch.

(d) Using the given basal heat flow condition for both, do you expect the new equilibrium surface heat flow to differ from the pre-event value in either scenario? Explain your reasoning before sketching the equilibrium curves.

(e) Sketch two or three intermediate geotherms for each scenario, showing the path from your initial curve to your equilibrium curve.

(f) In sedimentation, if the thermal conductivity of the fill is lower than the crust below—often a reasonable assumption, how would that affect the shape of the equilibrium geotherm?



Question 2. Sedimentation vs. Erosion

(a) On the Sedimentation panel, sketch the geotherm immediately after deposition.

(b) On the Erosion panel, sketch the geotherm immediately after erosion.

(c) In Sedimentation, is the resulting discontinuity a kink *within* the profile, or a jump *at* the surface? What about Erosion? Mark the point of highest curvature on each sketch.

(d) Using the given basal heat flow condition for both, do you expect the new equilibrium surface heat flow to differ from the pre-event value in either scenario? Explain your reasoning before sketching the equilibrium curves.

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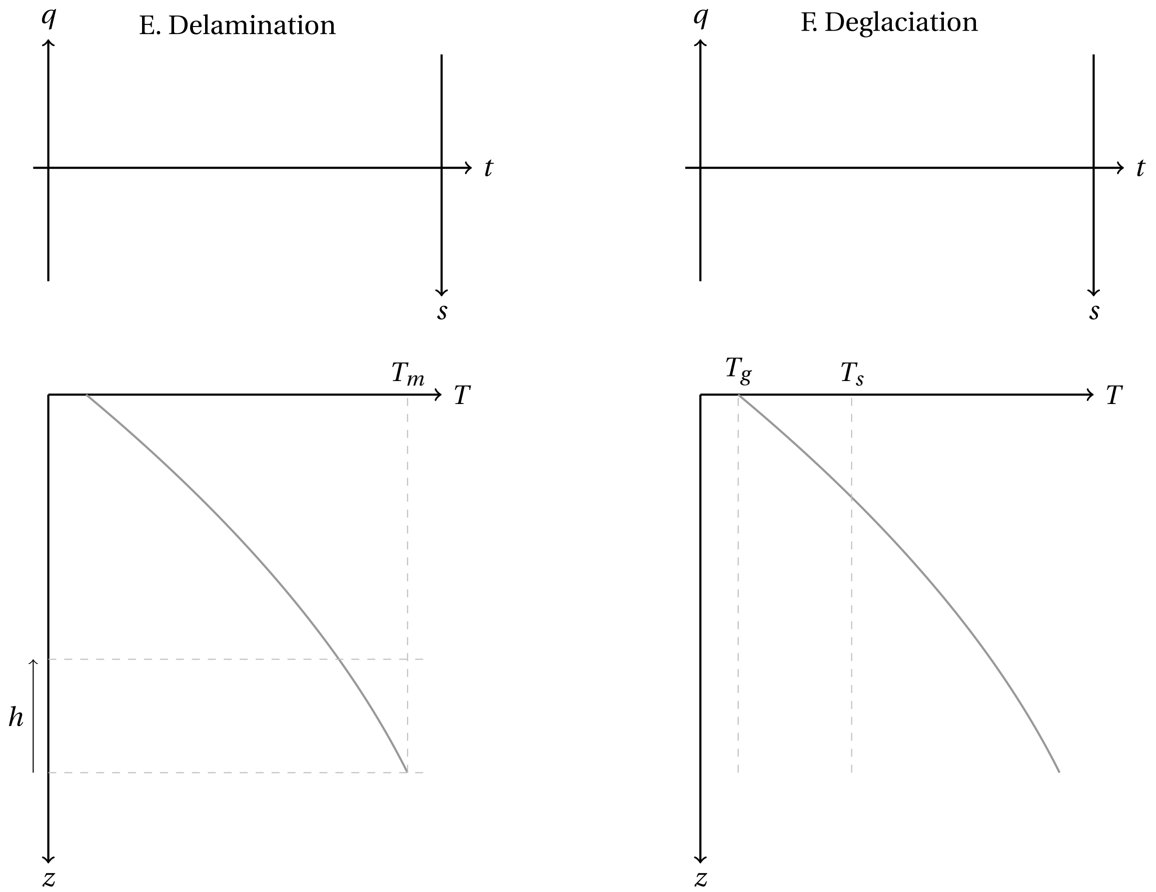

Question 3. Delamination vs. Deglaciation

(a) On the Delamination panel, sketch the geotherm immediately after foundering.

(b) On the Deglaciation panel, sketch the geotherm immediately after the temperature step.

(c) Both of these change a boundary condition rather than inserting a new layer – but Delamination changes the *deep* boundary and Deglaciation changes the *shallow* one. Which do you expect to change surface heat flow sooner, and why?

(d) Sketch two or three intermediate geotherms for each scenario, and sketch your predicted heat flow and subsidence curves on the upper axes for all six scenarios from this activity.

Key Ideas

A kink smooths fast; a uniform insulating layer changes the equilibrium slowly. These are not the same process happening at different speeds—they are genuinely different things. The sharp curvature at a fresh boundary relaxes quickly; the slower drift of the whole column toward its new equilibrium is a separate, much longer process governed by the whole column's response, not by that one point.

Adding or removing material with no heat production of its own changes temperatures without necessarily changing surface heat flow. Because nothing new is generating heat, the same flux has to get through.

Not every perturbation is conductive-only. Latent heat release (Intrusion) and a change in the deep boundary condition itself (Delamination) both push the system away from the simple "curvature predicts rate of change" rule that holds when $A = 0$ and nothing else is adding or removing heat along the way.

Activity 9.C
Thermal Diffusivity, Thermal Length, and Timescales

Purpose

To understand transient heat flow in terms of:

- how far thermal signals propagate in a given time,
- how long it takes thermal changes to affect a given depth,
- and how the *thermal length scale* controls signal shape and timing.

Learning Outcomes

By the end of this activity, you should be able to:

- Define thermal diffusivity and explain which material properties control it.
- Use the thermal length scale $\ell_T = 2\sqrt{\kappa t}$ to estimate how far a thermal signal has propagated or how long it takes to reach a given depth.
- Interpret the error function solution and identify where temperature changes are largest.
- Compare thermal diffusion timescales for different materials and connect these to real geological processes, including how long a thermal event at a given crustal or lithospheric depth takes to be felt elsewhere.

Box .2: Thermal Diffusion: Solution and Scaling

Half-Space Solution

For instantaneous heating or cooling at a boundary, the temperature evolution is given by the error function solution:

$$T(z,t) = T_s + (T_b - T_s)\,\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Important: the argument of the error function,

$$\eta = \frac{z}{2\sqrt{\kappa t}},$$

controls the *shape and timing* of the thermal signal: the temperature change is largest where η is small and becomes negligible once η is of order 1. That crossover, $\eta \sim 1$, is what defines the characteristic depth below.

Box .2: continued.

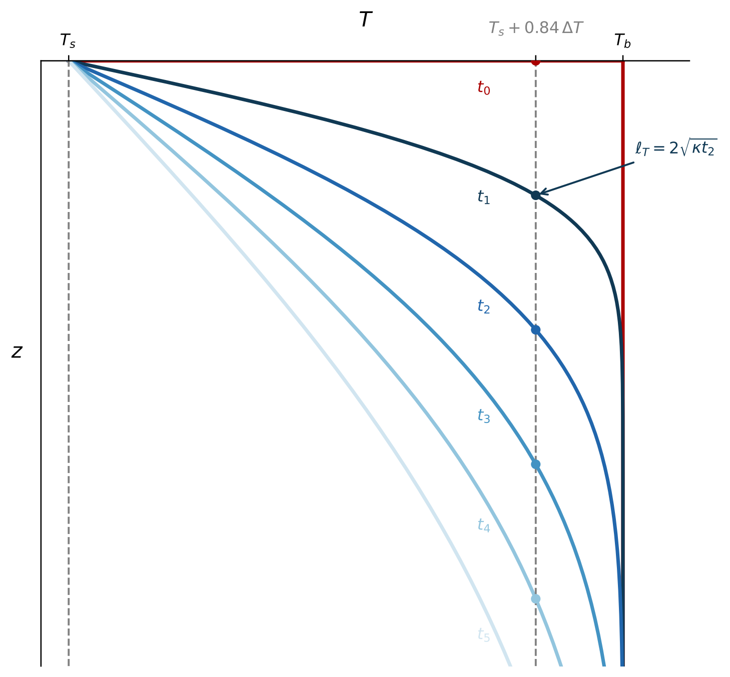

Figure 1: Half-space cooling at several time intervals. Each curve reaches $T_s + 0.84\,\Delta T$ (dashed line) at its own thermal length $\ell_T = 2\sqrt{\kappa t}$ – the same fractional temperature change, at a depth that grows with time.

Thermal Length and Characteristic Time

For diffusion-dominated heat transfer, the characteristic depth over which temperature changes propagate scales as

$$z \sim 2\sqrt{\kappa t}.$$

This thermal length sets both:

- the depth to which temperature changes have propagated after time t,
- the timescale required for a thermal perturbation at depth to influence shallower levels.

Thermal Diffusivity

Thermal diffusivity is defined as

$$\kappa = \frac{k}{\rho C_p},$$

where k is thermal conductivity, ρ is density, and C_p is specific heat.

Thermal conductivity and diffusivity vary with temperature, porosity, mineralogy, fluid content, and fabric. While not simple, there are some typical ranges that one can expect depending on the geologic material (Table **??**).

Table 1: Representative ranges for thermal conductivity and diffusivity of geologic materials. Values are rounded for scaling arguments rather than precise modeling.

Box .2: continued.

Material	Thermal Conductivity k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	Thermal Diffusivity κ ($\mathrm{m^2\,s^{-1}}$)	Thermal Diffusivity κ ($\mathrm{km^2\,My^{-1}}$)
Ice (glacial / ice sheet average)	1.6–2.0	$(0.7–0.9) \times 10^{-6}$	22–28
Clay-rich sediments	0.6–1.5	$(0.2–0.5) \times 10^{-6}$	6–16
Granite	2.5–3.5	$(0.8–1.2) \times 10^{-6}$	25–38
Basalt / Gabbro	1.7–2.2	$(0.6–1.0) \times 10^{-6}$	19–32
Peridotite (upper mantle)	3.3–4.5	$(1.0–1.5) \times 10^{-6}$	32–47

Tasks

Question 1. Rock vs. Ice

Using the diffusivity table, compute the thermal length $\ell_T = 2\sqrt{\kappa t}$ for rock (use granite, $\kappa \approx 30\ \mathrm{km^2/My}$) and for ice ($\kappa \approx 25\ \mathrm{km^2/My}$) at each timescale below. The first row is worked for you.

t (yr)	t (My)	ℓ_T, rock (km)	ℓ_T, ice (km)
10^3	10^{-3}	0.35	0.32
10^5			
10^7			

(a) Describe how the depth of thermal influence differs between rock and ice at each timescale.

(b) Explain why borehole paleothermometry in rock requires much greater depths to recover long-term climate signals than ice-core records – is it mainly a difference in κ, or something else?

Question 2. Distance Given Time

(a) For a fixed time t, describe how temperature varies with depth z as the value of η increases.

(b) Explain why temperature changes are largest for $\eta \ll 1$ and become negligible for $\eta \gg 1$.

(c) Using Figure **??**, how does the depth at which 84% of ΔT has been reached change between t_1, t_2, and t_3? Does it scale the way you'd expect from $\ell_T = 2\sqrt{\kappa t}$?

Question 3. Time Given Distance: How Long Does a Thermal Event Take to Be Felt?

Rearranging the thermal length relation,

$$t \sim \frac{z^2}{4\kappa},$$

gives the timescale for heat to travel a distance z. Using $\kappa \approx 30$ km^2/My (representative crust/mantle value), complete the table below. The first row is worked for you.

Depth	z (km)	t (My)
Mid-lower crust	20	3.3
Moho	35	
LAB	100	

(a) Consider a plume impinging on the base of the lithosphere. Using your table, roughly how long would it take before this thermal perturbation could influence surface temperatures?

(b) Following delamination of a dense crustal root, heat is supplied from below. Using your table, explain why crustal heating is delayed rather than instantaneous – and delayed by roughly how long?

Question 4. Interpreting "Arrival" of Heat

(a) Does heat arrive at a specific time, or does it diffuse gradually? Use the error function shape in Figure **??** to justify your answer.

(b) For $\eta \approx 1$, what fraction of the total temperature change has occurred? What does this imply about defining a single thermal "response time"?

(c) Discuss why the concept of a sharp thermal front is inappropriate for diffusion-dominated heat transport.

Geological Interpretation

- Thick lithosphere acts as a strong thermal filter, delaying surface response to basal heating.
- Thermal signals generated at depth are attenuated and spread out in time as they propagate upward.
- The thermal length scale controls which geological processes can be observed at the surface, and on what timescales.

Key Ideas

Diffusion has a characteristic relationship between time and space. In diffusion problems, time and distance are interchangeable through the thermal length $\ell_T = 2\sqrt{\kappa t}$. A thermal perturbation at the Moho takes millions of years to be felt at the surface; one at the base of the lithosphere takes tens of millions – diffusion is slow enough that most crustal thermal events are, for practical purposes, invisible at the surface on human timescales.

Depth and time trade off along straight lines, not a single front. Each value of $\eta = z/(2\sqrt{\kappa t})$ traces a straight line through the origin in a plot of z against $\sqrt{\kappa t}$, since $z = 2\eta\sqrt{\kappa t}$. A small η (strongly affected) and a large η (essentially unaffected) bound a widening wedge rather than a sharp edge, so "has heat arrived here yet?" only has an answer once you specify how much temperature change counts as arrival.

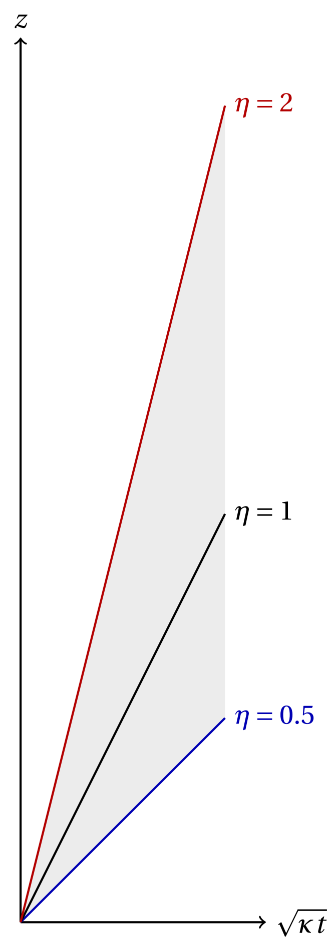

Figure 2: Depth z versus $\sqrt{\kappa t}$ at fixed $\eta = z/(2\sqrt{\kappa t})$. Each η traces a straight line of slope 2η; the shaded wedge between $\eta = 0.5$ and $\eta = 2$ spans the range over which the thermal response fades from strong to negligible, rather than switching off at a single depth.

Activity 9.D
Oceanic Lithosphere: Half-Space and Plate Cooling

Purpose

Newly formed oceanic lithosphere cools as it moves away from a ridge, and that cooling leaves two independent, measurable fingerprints: surface heat flow and seafloor depth. This activity derives both from the half-space cooling solution, shows why each becomes linear in a particular transformed variable, and tests the model against real global compilations—including where it breaks.

Learning Outcomes

By the end of this activity, you should be able to:

- Derive surface heat flow and subsidence from the half-space cooling solution $T(z,t)$.
- Explain why q^{-2} is linear in age and why depth is linear in $\sqrt{\text{age}}$, for a half-space.
- Identify where real oceanic data does and does not follow this prediction, and connect any departure to the plate cooling model's finite basal boundary condition.
- Relate the timescale at which half-space and plate predictions diverge to the characteristic time $\tau = L^2/\kappa$.

Background

The half-space cooling solution, with $T(0,t) = T_0$ and $T(z,0) = T_m$ for all $z > 0$, is

$$T(z,t) = T_0 + (T_m - T_0)\,\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

This is the same error-function half-space solution used for any instantaneous boundary-temperature change, now applied with T_0 as the seafloor (bottom water) temperature and T_m as the mantle temperature it cools from.

Aside: oceanic lithosphere technically forms by seafloor spreading, the $\gamma = 1$ limit of the rifting model in your notes and problem set—that limit is singular in the pure-shear formulation and requires a separate treatment, given in your notes. We do not need it here; the half-space and plate solutions below are self-contained.

Tasks

Question 1. Deriving Surface Heat Flow

Using $q_0(t) = k\left.\dfrac{\partial T}{\partial z}\right|_{z=0}$ and the chain rule, along with $\dfrac{d}{dx}\mathrm{erf}(x) = \dfrac{2}{\sqrt{\pi}}e^{-x^2}$, derive an expression for $q_0(t)$.

You should arrive at $q_0(t) = \dfrac{k(T_m - T_0)}{\sqrt{\pi\kappa t}}$.

Question 2. Deriving Subsidence

Thermal contraction of the cooling column, balanced isostatically against the water that fills the space it vacates, produces subsidence

$$s(t) = \frac{\rho_m}{\rho_m - \rho_w}\,\alpha \int_0^\infty \big[T(z,0) - T(z,t)\big]\,dz.$$

(a) Write $T(z,0) - T(z,t)$ in terms of the complementary error function, $\operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta)$.

(b) Substitute $\eta = z/(2\sqrt{\kappa t})$ to change variables in the integral. You will need the identity $\int_0^\infty \operatorname{erfc}(\eta)\,d\eta = \dfrac{1}{\sqrt{\pi}}$ (given; this one isn't worth re-deriving here).

(c) Combine your results to arrive at $s(t)$.

You should arrive at $s(t) = \dfrac{2\rho_m\alpha(T_m - T_0)}{\rho_m - \rho_w}\sqrt{\dfrac{\kappa t}{\pi}}$.

Question 3. Why These Linearize

(a) Using your result from Question 1, show that q_0^{-2} is linear in t.

(b) Using your result from Question 2, in what variable is $s(t)$ linear?

Question 4. Estimating the Fit Constants by Hand

Write the half-space predictions as $q(t) = c_q/\sqrt{t}$ and $s(t) = c_s\sqrt{t}$, absorbing all the physical constants (k, κ, $T_m - T_0$, ρ_m, α, $\ldots$) into the single constants c_q and c_s. Two real data points from each dataset are given below; s is depth relative to the ridge-depth reference of 2500 m.

Heat flow	
t (Ma)	q (mW m^{-2})
19.8	114.0
51.3	72.1

Bathymetry		
t (Ma)	depth (m)	s =depth−2500 (m)
11.3	3633	
41.3	4633	

(a) Using $q(t) = c_q/\sqrt{t}$, compute c_q from each heat flow point separately. Are your two estimates consistent with each other?

(b) Using $s(t) = c_s\sqrt{t}$, compute c_s from each bathymetry point separately. Are your two estimates consistent with each other?

(c) Which pair of estimates agreed more closely – c_q or c_s? Given what generates scatter in young oceanic heat flow, suggest why one dataset might let a small number of points pin down its constant more reliably than the other.

Question 5. Real Data: Heat Flow

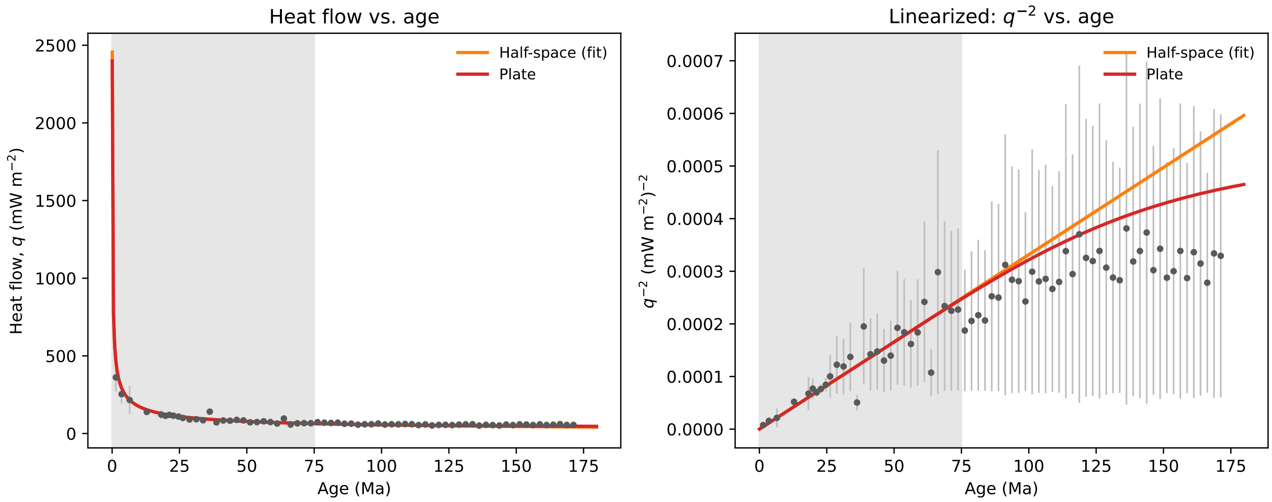

Figure 3: Sediment-corrected global heat flow vs. seafloor age (binned, error bars show one standard deviation). The half-space curve here is fit by least squares to *every* point in the shaded region ($t < 75$ Ma), giving $c_q \approx 549$ mW m^{-2} Ma$^{1/2}$.

(a) How does the full-dataset fit, $c_q \approx 549$, compare to your two hand estimates from Question 4? Does this change how much you'd trust a constant estimated from just one or two points?

(b) At young ages, does the data sit above, below, or on the half-space prediction? What physical process, not included in this conductive model, might explain that?

(c) At the oldest ages shown, do half-space and plate predictions still agree closely, or have they begun to separate? What does that tell you about how old the seafloor needs to be before the two models are distinguishable in heat flow alone?

Question 6. Real Data: Bathymetry

(a) How does this full-dataset fit compare to your two hand estimates from Question 4? Is the

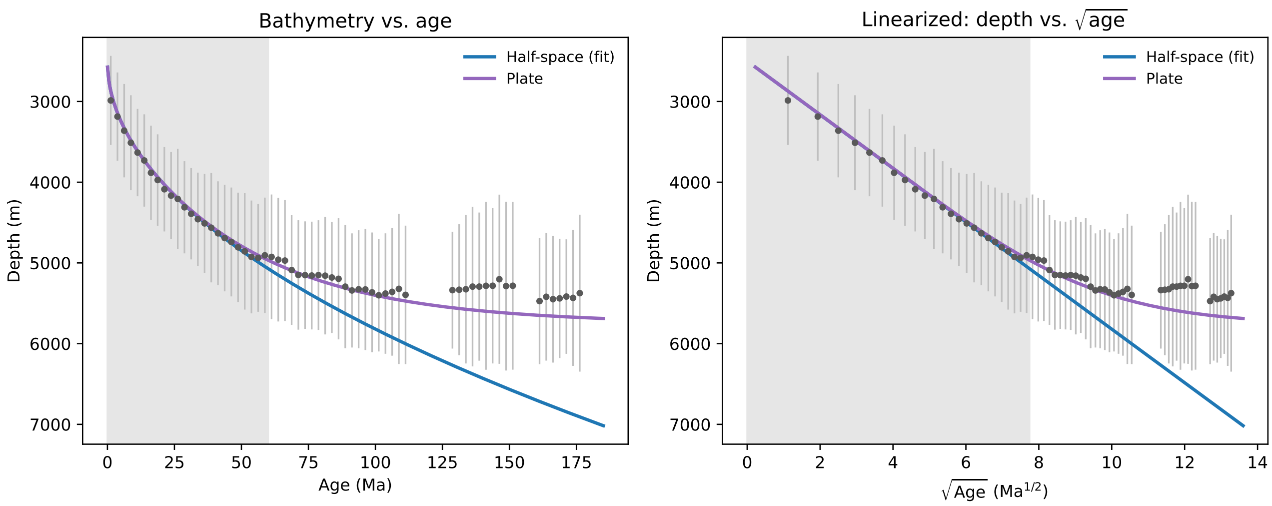

Figure 4: Global bathymetry vs. seafloor age (binned, error bars show one standard deviation). The half-space curve is fit to every point in the shaded region ($t < 60$ Ma), giving $c_s \approx 332\ \mathrm{m\,Ma^{-1/2}}$.

improvement from using more points as large as it was for heat flow?

(b) Compare this figure to Question 5's heat flow figure. In which observable does the half-space model's failure become obvious sooner, as a function of age?

(c) The half-space model has no plate thickness L in it at all—it's a model of infinite depth. Explain, physically, why real subsidence eventually stops following the half-space prediction.

Question 7. Early and Late Time

The characteristic time for the plate model is $\tau = L^2/\kappa$.

(a) For $t \ll \tau$, the plate solution reduces to the half-space solution. Why does a finite-thickness plate behave like an infinite half-space when it's young?

(b) For $t \gg \tau$, heat flow approaches a constant, $q_0 \to k(T_m - T_0)/L$, independent of t. Why does the plate's finite thickness impose a floor on how thin the thermal boundary layer can get, when a half-space's doesn't?

Key Ideas

The same solution predicts two independent, testable observables. Heat flow comes from the temperature gradient at the surface; subsidence comes from integrating the temperature deficit through the whole column. Both derive from the same $T(z,t)$, but they linearize in different variables (t vs. $\sqrt{t}$) and, as you saw, they reveal the half-space model's breakdown at different ages.

A model with no built-in length scale cannot predict its own failure. The half-space solution has no plate thickness in it, so nothing internal to the model signals when it stops applying, that only becomes visible by comparing to real data, or to a model (plate cooling) that does have a length scale.

Early-time agreement between two models doesn't guarantee late-time agreement. The plate and half-space solutions coincide for $t \ll \tau$ by construction, not by coincidence—and diverge once the finite basal boundary condition is actually felt, at $t \sim \tau$.